

EE 316

Section 3

Pre-Lab 01

By: Hannah Kelley

Introduction:

The purpose of this lab is to review concepts used to analyze basic circuits such as mesh current analysis, nodal analysis, and Thevenin and Norton equivalent circuits. This lab also serves as a hands-on way to validate basic circuit laws such as Ohm's Law Kirchhoff's Current and Voltage Laws.

Simulations:

Part 1A

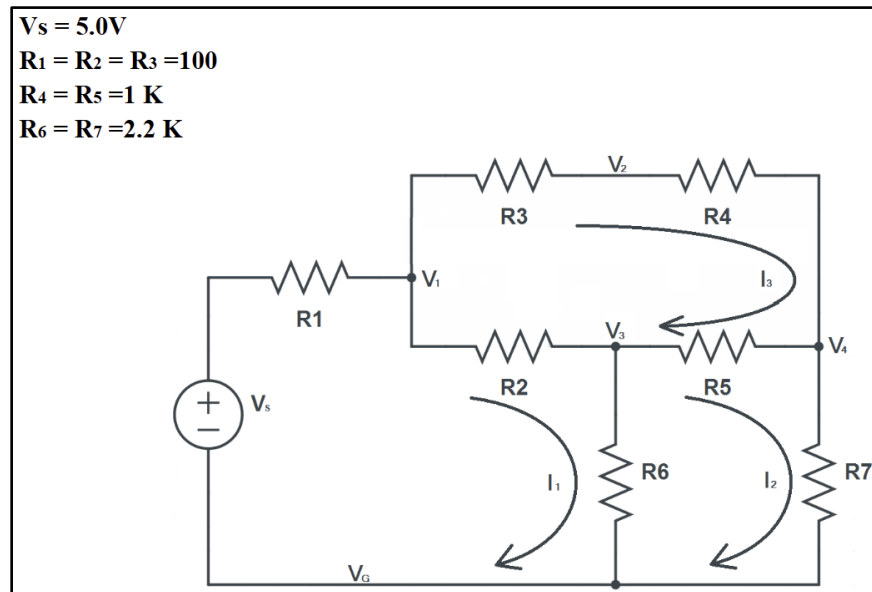


Figure #1: Circuit from Lab Manual (Part A)

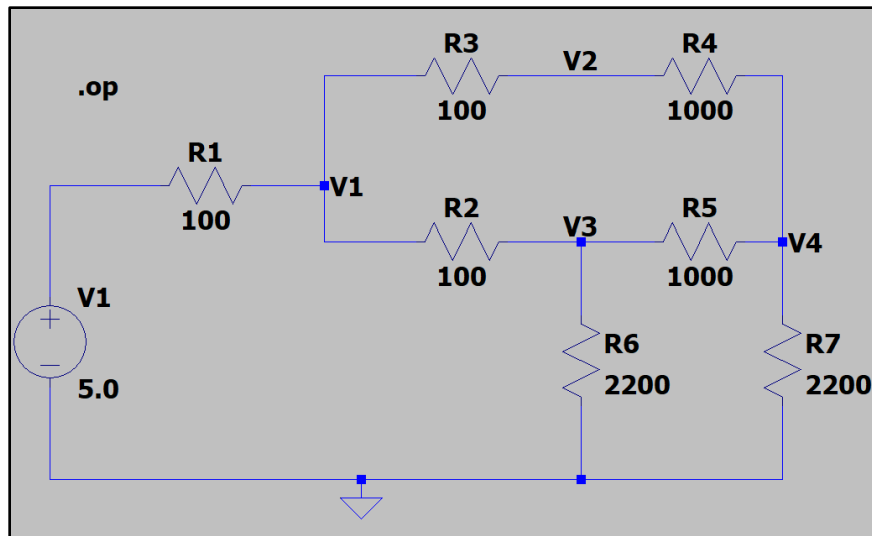


Figure #2: Simulation in LT Spice (Part A)

```

* C:\Users\hanna\OneDrive\Documents\LTspiceXVII\prelab1_part1a.asc

--- Operating Point ---

V(n001) :      5          voltage
V(v1) :      4.63661     voltage
V(v3) :      4.36473     voltage
V(v2) :      4.54509     voltage
V(v4) :      3.62993     voltage
I(R7) :      0.00164997   device_current
I(R6) :      0.00198397   device_current
I(R5) :      0.000734803   device_current
I(R4) :      0.000915164   device_current
I(R3) :      0.000915164   device_current
I(R2) :      0.00271877   device_current
I(R1) :      0.00363393   device_current
I(V1) :      -0.00363393   device_current

```

Figure #3: Simulation Output from LT Spice (Part A)

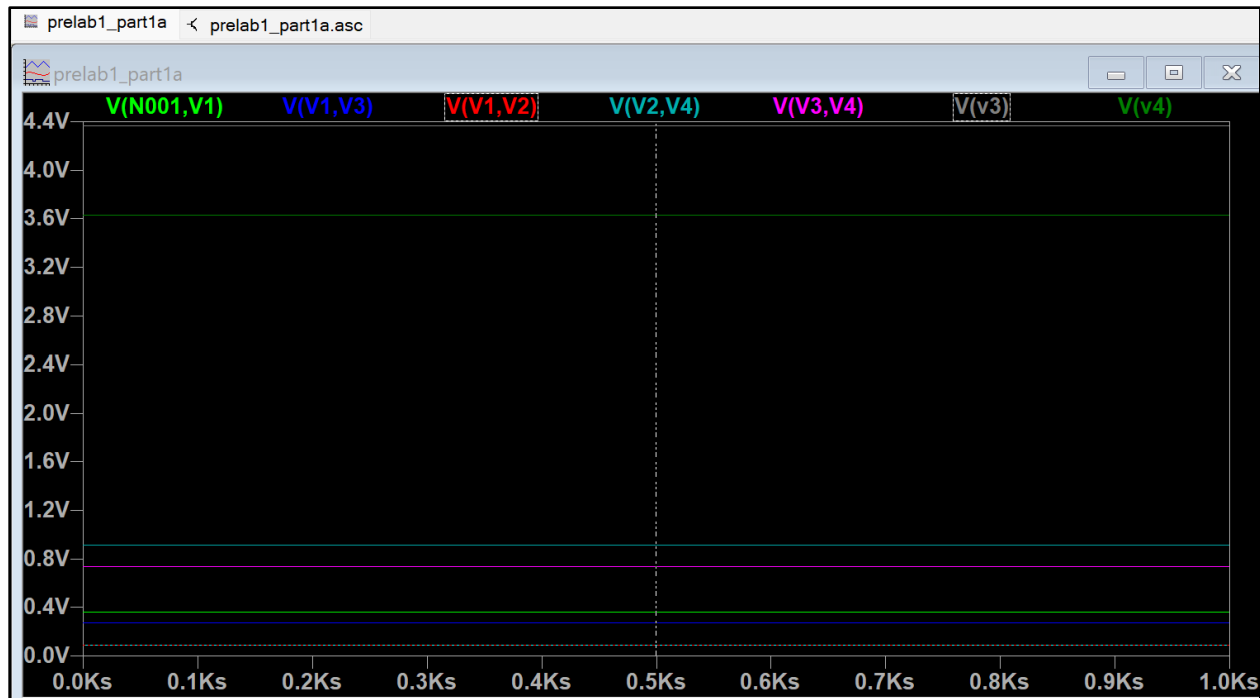


Figure #4: Branch Voltage Data from LT Spice Simulation (Part A)

Branch Voltages	Branch Currents	Node Voltages	Loop Currents
$V_{R1} = 363.39331 \text{ mV}$	$i_{R1} = 3.63393 \text{ mA}$	$V_1 = 4.63661 \text{ V}$	$I_1 = 3.63393 \text{ mA}$
$V_{R2} = 271.87729 \text{ mV}$	$i_{R2} = 2.71877 \text{ mA}$	$V_2 = 4.54509 \text{ V}$	$I_2 = 1.64997 \text{ mA}$
$V_{R3} = 91.516495 \text{ mV}$	$i_{R3} = 0.915164 \text{ mA}$	$V_3 = 4.36473 \text{ V}$	$I_3 = 0.915164 \text{ mA}$
$V_{R4} = 915.16376 \text{ mV}$	$i_{R4} = 0.915164 \text{ mA}$	$V_4 = 3.62993 \text{ V}$	
$V_{R5} = 734.80296 \text{ mV}$	$i_{R5} = 0.734803 \text{ mA}$		
$V_{R6} = 4.3647294 \text{ V}$	$i_{R6} = 1.98397 \text{ mA}$		
$V_{R7} = 3.6299264 \text{ V}$	$i_{R7} = 1.64997 \text{ mA}$		

Table #1: Simulation Data for Part A

Part 1B

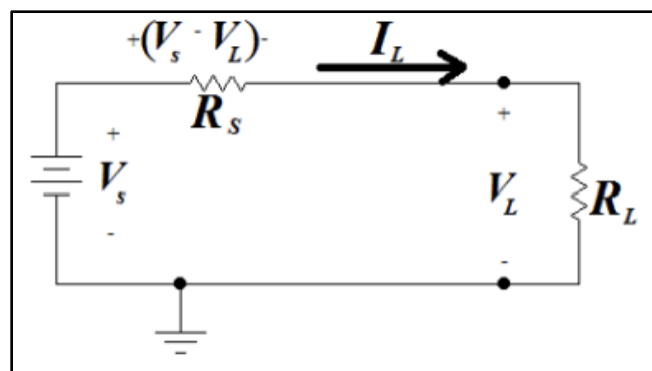


Figure #5: Circuit from Lab Manual (Part B)

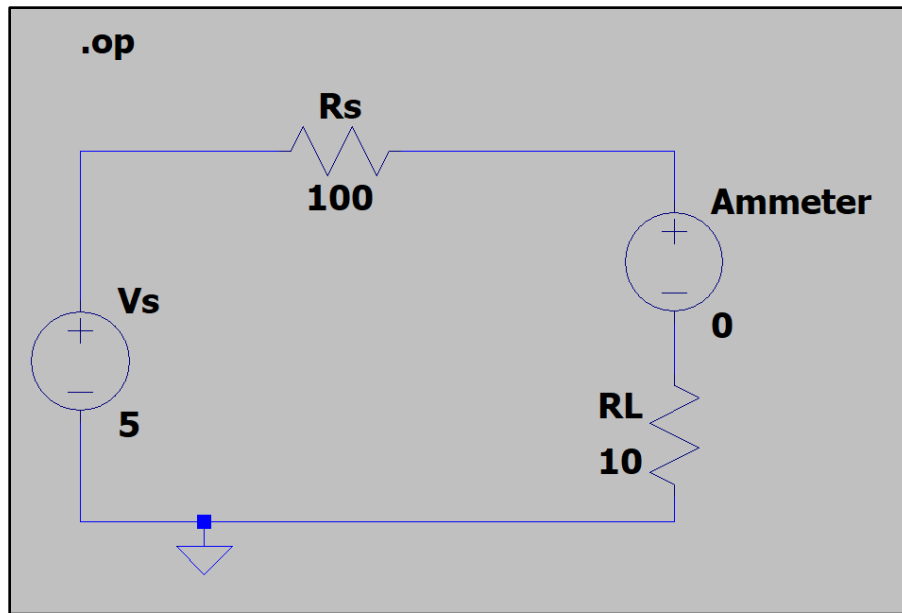


Figure #6: Simulation in LT Spice at $R_L = 10$ Ohms (Part B)

```

--- Operating Point ---
V(n002) :      0.454545      voltage
V(n001) :      5            voltage
V(n003) :      0.454545      voltage
I(RL) :       -0.0454545     device_current
I(Rs) :       -0.0454545     device_current
I(Ammeter) :    0.0454545     device_current
I(Vs) :       -0.0454545     device_current

```

Figure #7: Output of Part B Simulation at $R_L = 10$ Ohms

R_L (Ohms)	V_L (V)	I_L (mA)
10	454.54547 mV	45.454547 mA
20	833.33331 mV	41.666668 mA
40	1.4285715 V	35.714287mA
70	2.0588236 V	29.411765mA
100	2.5 V	25 mA
150	3 V	20 mA
200	3.3333333 V	16.666668 mA
300	3.75 V	12.5 mA
500	4.1666665 V	8.3333338 mA
1000	4.5454545 V	4.5454544 mA

Table #2: Simulation Data for Part B

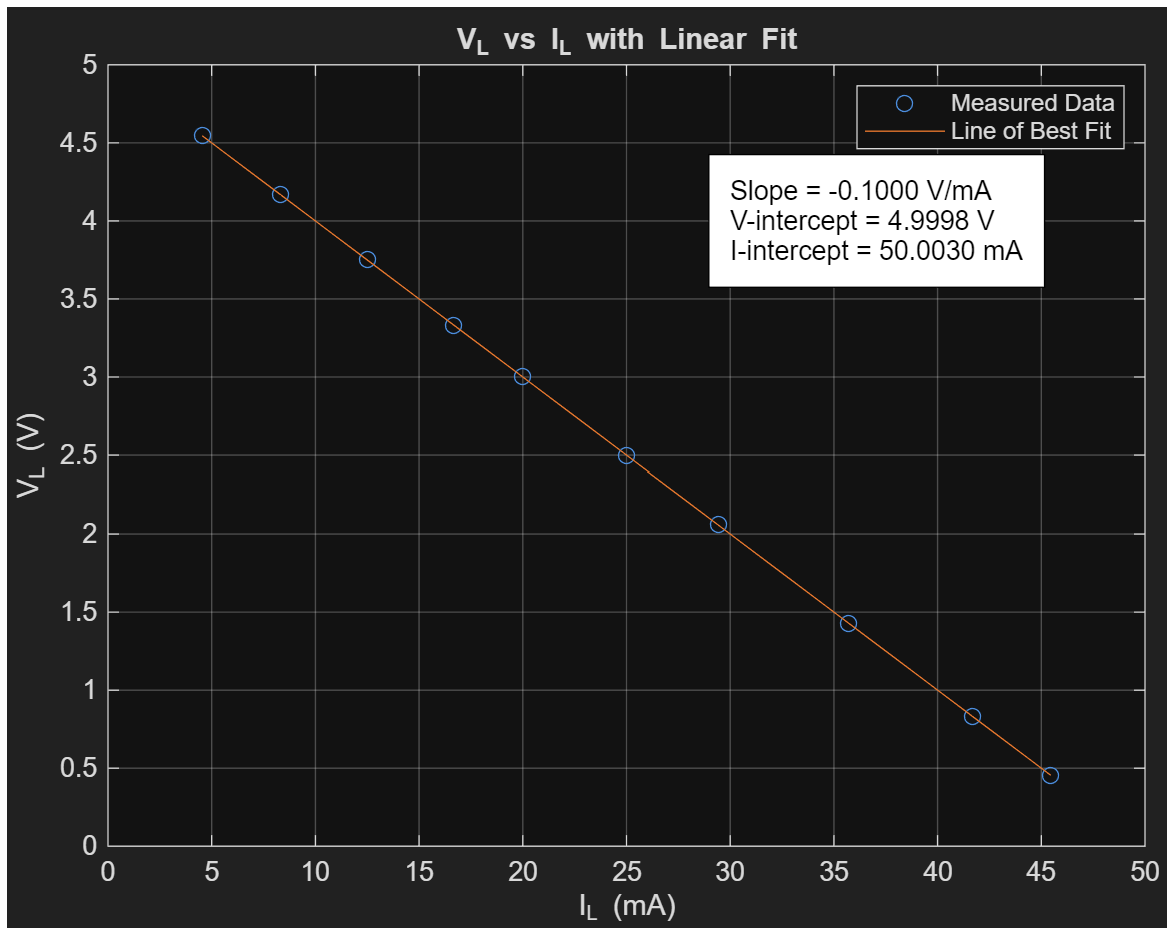


Figure #8: Load Voltage vs. Load Current (MATLAB)

```
IL = [45.455 41.667 35.714 29.412 25 20 16.667 12.5 8.333 4.545]; % in Amps
```

```
VL = [0.455 0.833 1.429 2.059 2.5 3 3.333 3.75 4.167 4.545]; % in Volts
```

```
p = polyfit(IL, VL, 1); % eq for line of best fit
```

```
VL_fit = polyval(p, IL);
```

```
slope = p(1);
```

```
y_intercept = p(2);
```

```
x_intercept = -y_intercept / slope;
```

```
plot(IL, VL, 'o')
```

```
hold on
```

```
plot(IL, VL_fit)
```

```
hold off
```

```
xlabel('I_L (mA)');
```

```
ylabel('V_L (V)');
```

```
title('V_L vs I_L with Linear Fit');
```

```
legend('Measured Data', 'Line of Best Fit', 'Location', 'northeast');
```

```
grid on;
```

```

text(30, 4.0, sprintf(['Slope = %.4f V/mA\n' ...
    'V-intercept = %.4f V\n' ...
    'I-intercept = %.4f mA'], ...
    slope, y_intercept, x_intercept), ...
    'Color', 'black', ...
    'BackgroundColor', 'white', ...
    'EdgeColor', 'black', ...
    'Margin', 8);

```

Table #3: MATLAB Code Used to Produce Figure 8

Part 2

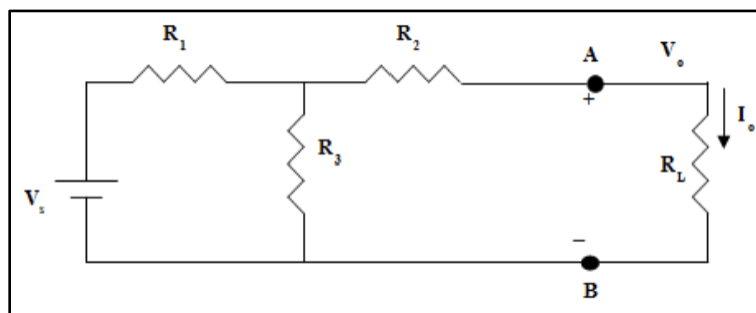


Figure #9: Circuit from Lab Manual (Part 2)

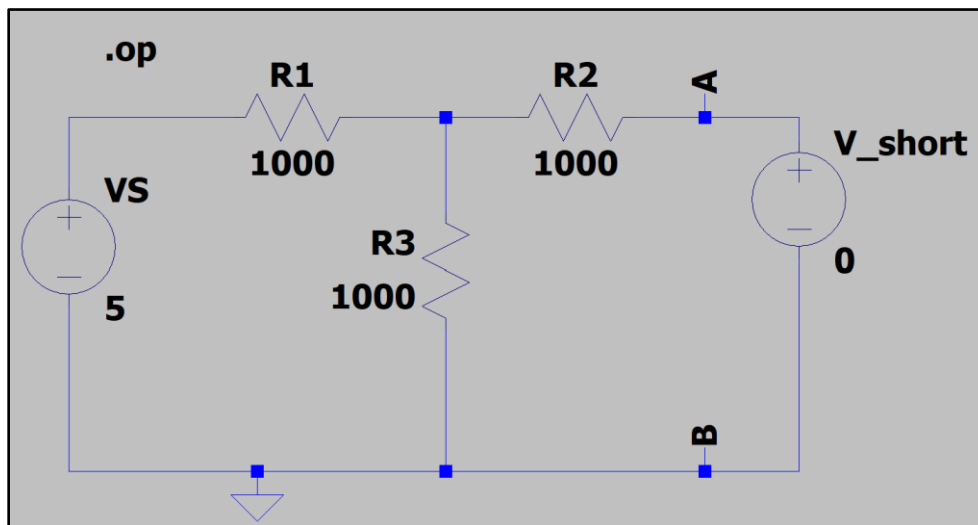


Figure #10: Simulated Circuit for Part 2 (V_{short})

* C:\Users\hanna\OneDrive\Documents\LTspiceXVII\part2_vshort.asc		
--- Operating Point ---		
V(n002) :	1.66667	voltage
V(n001) :	5	voltage
V(a) :	0	voltage
I(R3) :	-0.00166667	device_current
I(R2) :	-0.00166667	device_current
I(R1) :	-0.00333333	device_current
I(V_short) :	0.00166667	device_current
I(Vs) :	-0.00333333	device_current

Figure #11: Simulated Outputs of V_short

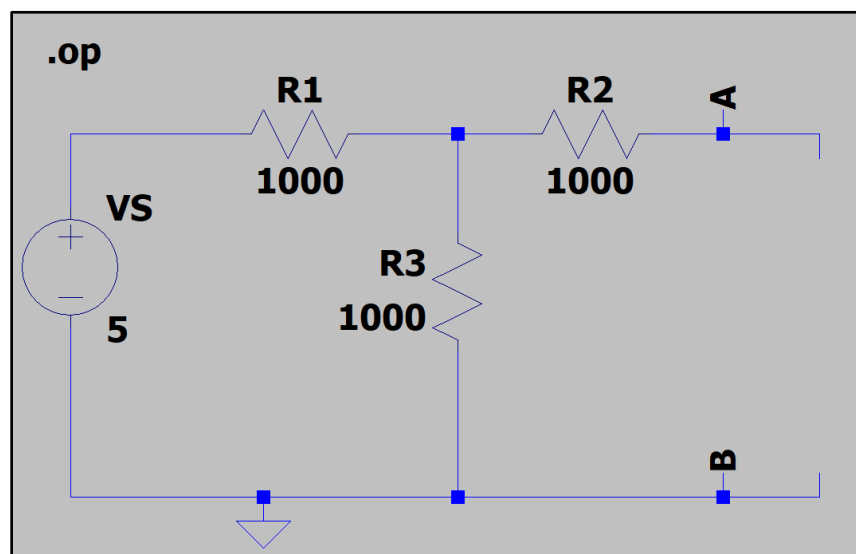


Figure #12: Simulated Circuit for Part 2 (I_{oc})

* C:\Users\hanna\OneDrive\Documents\LTspiceXVII\part2_vshort.asc		
--- Operating Point ---		
V(n002) :	2.5	voltage
V(n001) :	5	voltage
V(a) :	2.5	voltage
I(R3) :	-0.0025	device_current
I(R2) :	0	device_current
I(R1) :	-0.0025	device_current
I(Vs) :	-0.0025	device_current

Figure #13: Simulated Outputs of I_{oc}

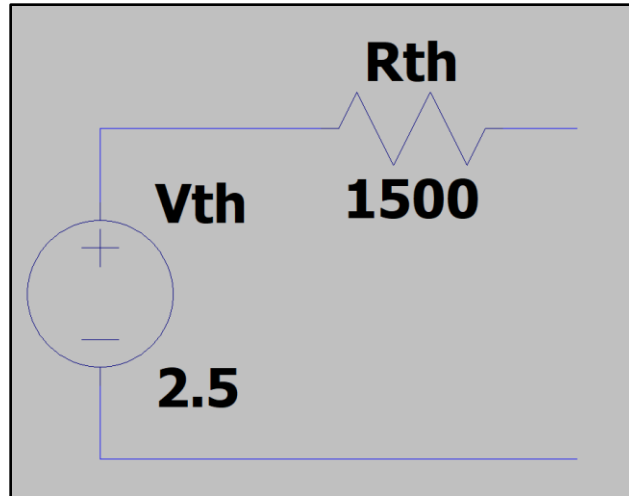


Figure #14: Thevenin Equivalent Circuit

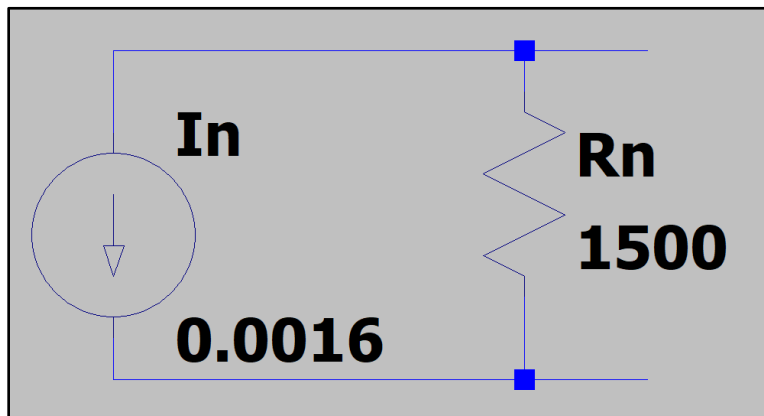


Figure #15: Norton Equivalent Circuit

Circuit	R _L (Ohms)	P _{max}
Thevenin	1500	$P_{max} = \frac{V_{th}^2 R_L}{2R_{TH}} = 1.042 \text{ mW}$
Norton	1500	$P_{max} = \frac{I_N^2 R_N^2 R_L}{(R_N + R_L)^2} = 0.960 \text{ mW}$

Table #4: Power Calculations of Thevenin and Norton Circuits